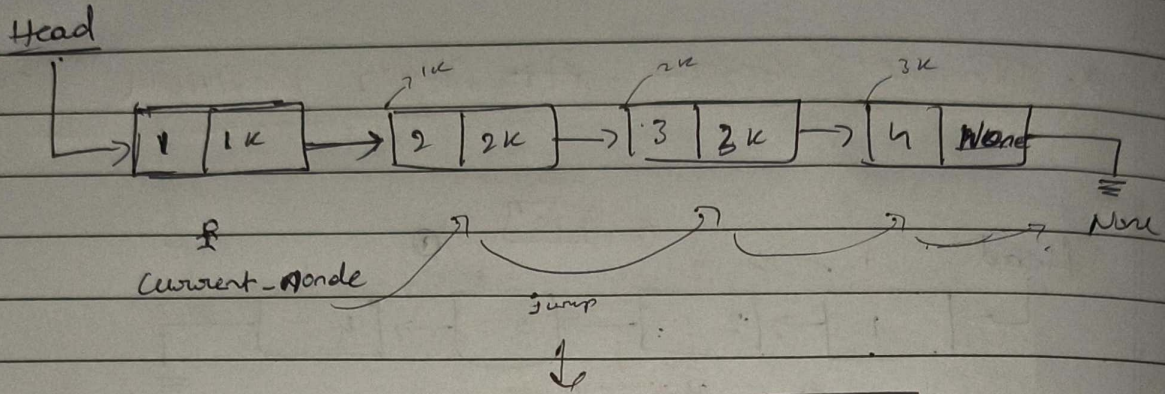


B Singly linked list in python - Search

Print all nodes & Search for a given Value

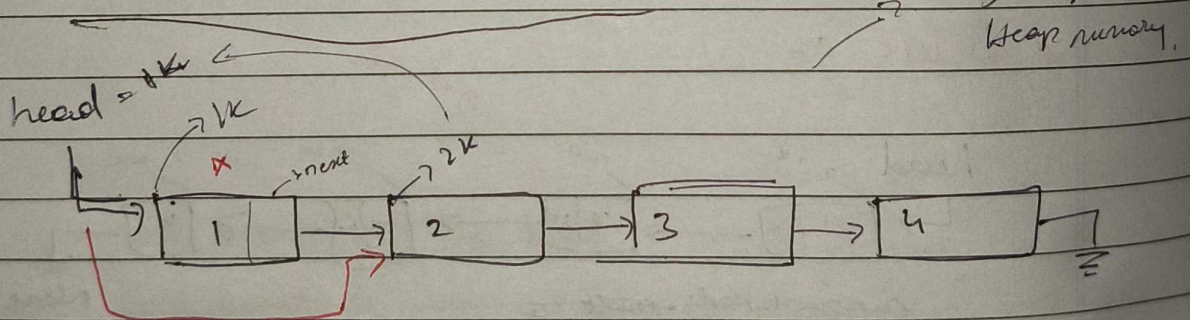


`Current-node = Current-node.next`

`data == key`

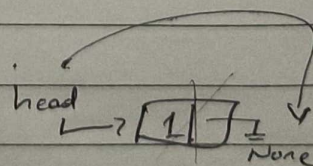
B Singly linked list in python - Delete

* Delete node start / End

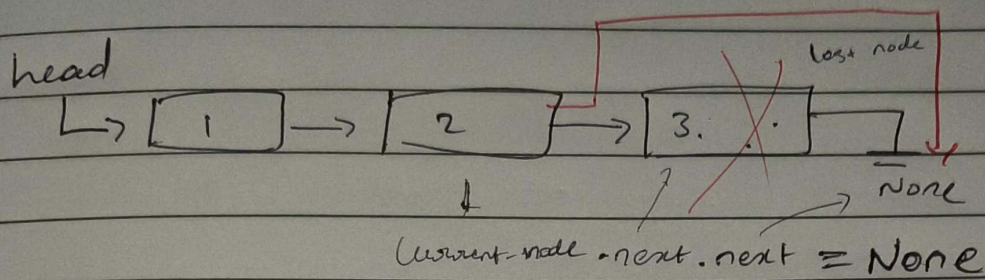


`head = head.next`

- ① Empty
- ② 1 node - only
- ③ 2 or more nodes

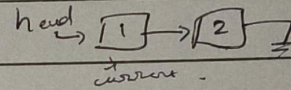


* delete node at End.



- ① Empty
- ② 1 node only
- ③ 2 or more nodes.

for only 2 nodes.



While $\text{current-node.next.next} \neq \text{None}$:

```
def delete_at_beginning(self):  
    if self.head is None:  
        print("list is empty")  
        return  
  
    self.head = self.head.next
```


Structure in Python / 02_Singly linked list / 4 Delete at start and

```
class SinglyLinkedList:
```

```
    def delete_at_end(self):
```

```
        if self.head is None:
```

```
            print("list is empty")
```

```
            return
```

```
        if self.head.next is None:
```

```
            self.head = None
```

```
            return
```

```
        current_node = self.head
```

```
        while current_node.next.next is not None:
```

```
            current_node = current_node.next
```

```
        current_node.next = None
```



```
class SinglyLinkedList:
    def search(self, key):

        # when list is empty
        if self.head is None:
            print("List is empty, nothing to search.")
            return

        # start from head/currently in head node
        current_node = self.head

        # traverse the list to find the key
        while current_node is not None:
            if current_node.data == key:
                print(f"Value {key} found in the list.")
                return
            current_node = current_node.next
```